DIVINE GRACE COLLEGE OF ARTS, SCIENCE AND TECHNOLOGY (DGCAST) – GAROUA

GEOGRAPHY REVISION SHEET – FORM FOUR

First Sequence Evaluation, School Year 2026-2027 • Teacher: Mr KAMDEM Emmanuel

Lessons revised: Physical Geography, Lessons 1 to 30 (the Earth in space, structure, plates, weathering, rivers, waves, deserts, atmosphere)

A. THE EARTH IN SPACE (LESSONS 1 TO 6)

1. Time: how many degrees correspond to one hour? What is standard time?

Answer: 360° in 24 hours: **15° = 1 hour, 1° = 4 minutes**; places in the east are ahead. Time zones are belts of 15°; Cameroon uses GMT + 1. Rotation (west to east, 24 h) gives day and night, time differences, deflection of winds and tides; revolution (365¼ days) gives seasons, varying day length and leap year.

B. STRUCTURE OF THE EARTH, ROCKS AND PLATES (LESSONS 7 TO 13)

2. Describe the internal structure of the Earth.

Answer: Crust (5–10 km under oceans, 30–70 km under continents); **mantle** (about 2,900 km, very hot rock); **outer core** (about 2,200 km, liquid iron and nickel); **inner core** (solid iron and nickel, 5,000–6,000 °C).

3. Name the three types of rocks, their formation and an example of each.

Answer: Igneous: magma or lava cools (granite, basalt). **Sedimentary**: sediments laid in layers and pressed (sandstone, limestone, coal). **Metamorphic**: rocks changed by heat and pressure (marble, slate).

4. Continental drift and plate tectonics: main ideas and evidence.

Answer: Wegener (1912): one super-continent, **Pangaea**, broke up (Laurasia and Gondwanaland). Evidence: jigsaw fit of Africa and South America, same fossils, same rocks and mountains, ice marks in hot lands. Plate tectonics: 7 major plates moving 2–10 cm a year; margins: **constructive** (plates move apart: ridges, rift valleys), **destructive** (plates meet: fold mountains, trenches, volcanoes), **conservative** (plates slide: faults).

5. Volcanoes: types, structure, importance and hazards.

Answer: Active (Mount Cameroon), dormant, extinct (Manengouba); shield, composite and cinder cones. Parts: magma chamber, vent, cone, crater. Benefits: fertile soils, geothermal energy, tourism, building materials. Hazards: lava flows, ash, poisonous gases (Lake Nyos 1986), lahars. Measures: monitoring, evacuation, land-use planning, degassing.

6. Earthquakes: focus, epicentre, causes, effects and measures.

Answer: The **focus** is where it starts; the **epicentre** is the point on the surface above it. Causes: plate movements, eruptions, collapses, human activities. Measured with a seismograph (Richter scale 1–9). Effects: loss of life, destroyed property, fires, tsunamis. Measures: strong buildings, planning, warning, drills.

C. DENUDATION AND WEATHERING (LESSONS 14 TO 17)

7. Define denudation and weathering. Name the types and processes of weathering.

Answer: Denudation: wearing down of the land by weathering, erosion and transportation (agents: running water, wind, waves). **Physical weathering**: thermal shattering (exfoliation), frost shattering (scree), wetting and drying, pressure release. **Chemical weathering**: solution, carbonation (limestone caves), hydrolysis, oxidation (red soils), hydration.

D. RIVERS AND FLOODPLAINS (LESSONS 18 TO 22)

8. Describe the three courses of a river and their landforms.

Answer: Upper: steep, vertical erosion; V-shaped valley, interlocking spurs, waterfall, gorge, rapids, potholes. Middle: gentle slope, lateral erosion; meanders, river cliff, slip-off slope. Lower: almost flat, deposition; floodplain, levees, ox-bow lake, delta.

9. Give the mechanisms of river erosion and the ways a river carries its load. Man and the floodplain?

Answer: Erosion: hydraulic action, abrasion, attrition, solution. Transport: traction, saltation, suspension, solution. Floodplains give fertile land, water, fishing, energy and transport; problems: floods, diseases, displacement; solutions: dykes, dams, warnings, safe building.

E. WAVES, COASTS AND HOT DESERTS (LESSONS 23 TO 29)

10. Waves: types, erosional and depositional features.

Answer: Constructive waves (gentle, strong swash) build beaches; **destructive** waves (high, strong backwash) erode. Erosion: hydraulic action, abrasion, attrition, solution; features: cape, bay, cliff and wave-cut platform, cave, arch, stack, stump. Deposition by **longshore drift**: beach, **spit**, bar, tombolo.

11. Coasts and hot deserts: resources, hazards and management; wind features.

Answer: Coasts: ports, fishing, tourism, oil; hazards: erosion, floods, sea-level rise; management: sea walls, groynes, mangroves. Deserts (below 250 mm of rain): Sahara, Namib, Kalahari, Sahel; wind: deflection, abrasion, attrition; rock pedestals, zeugen, yardangs; dunes: barchans and seif.

F. THE ATMOSPHERE (LESSON 30)

12. Name the layers of the atmosphere. What are the causes and effects of ozone depletion?

Answer: Troposphere (0–12 km, weather occurs; temperature falls 6.5 °C per 1,000 m), stratosphere (12–50 km, ozone layer), mesosphere (50–85 km, coldest), thermosphere. Ozone is destroyed by CFCs, halons and other chemicals; more ultraviolet rays cause skin cancer and eye diseases.

Quick quiz	Answer
15° of longitude =	**1 hour**
Liquid layer of the Earth	**Outer core**

Plates move apart at …	**Constructive margins**
Strip of sand joined to the land at one end	**Spit**

Exam tip: *The Form 4 paper has* ***10 multiple choice questions*** *(0.5 mark each). Read all four options before choosing, and revise the* ***definitions*** *of each lesson: most questions come from them.*